

MOBILE SUIT GUNDAM: Silver Phantom

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Abstract

Bandai Namco Filmworks is pleased to announce that it will produce "Mobile Suit Gundam: Silver Phantom" the latest pioneering VR film based on the "Mobile Suit Gundam" series, with Atlas V, a VR production company that has created numerous immersive experiences, including "Gloomy Eyes" and "Battlescar." "Mobile Suit Gundam: Silver Phantom" is a visual work that offers fans the opportunity to immerse themselves in the Gundam world in a new way using VR and invites them into an unprecedented interactive story. This work will be distributed by Astrea, a company that has developed numerous VR titles, and will be available on Meta's VR headset, Meta Quest. Through this new VR experience of the Gundam world, we will deliver the appeal of Gundam to fans around the world.

CCS Concepts

• **Computing methodologies** → Virtual reality.

Keywords

Virtual reality, Modeling, Animation, Shader, Lighting, rendering, sakuga, storyboard

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1 Introduction

As an innovative VR film, this project reproduces the Japanese animation look on VR, and also develops challenging technologies that expand the possibilities of animation production expression and VR.

The narrative video length is approximately 60 minutes, and the interactive part is more than 20 minutes, totaling approximately 90 minutes of feature-length content. The content is not simply a 3Dof 360-degree video, but mostly 6Dof interactive real-time polygonal rendering, making the most of Meta Quest's superior performance.

Pre-production, mainly scenario and storyboarding, and CG modeling and animation were handled by Bandai Namco Filmworks of Japan. The interactive parts and lighting, effects and optimization were handled by ALBYON in France.

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This project uses 3dsMAX and Unity as the main 3DCG software tools. For modeling, 3dsMAX is used up to shape and rigging, and Unity for subsequent shaders and optimization. Animation was also done using 3dsMAX, and characters were created based on motion capture. Photoshop and Aftereffects were also used during the process to create the monitor graphics materials.

In addition, dedicated scripts were developed for exporting data from 3dsMAX and importing into Unity. The following is an overview of the process.

- 3DS Max Export script / Data and date check / Overall Sanity check / Camera check / Hierarchy check
- Hierarchy restructure and optimization / Animation prebaking / Split tracks / Publishing / Extra log info
- Batch process / Asset processing
- Unity Import script / Auto create levels / Populate with prefabs / Populate timeline and assign anime tracks
- Camera integration / VFX integration / Extract animations in Unity format / Optimize animation tracks
- Fix animation issues / Process SFX / Process dialogs / Process Subtitles / Batch process

2 VR development approach from traditional Japanese drawn animations

In order to incorporate animation production techniques for VR video, a new format of storyboard paper was developed instead of traditional animation storyboards. This format has six frames (front, back, top, bottom, left, and right), and is intended to convey the directorial elements of each frame without omission. Based on this storyboard, CG pre-visualization was created using the six-panel diagram, which was also utilized during sound production for the new immersive sound system, Ambisonics.

By incorporating the cut layout of the animation direction into the VR images, we discovered new possibilities in VR storytelling, which until now had been limited to the subjective.

And one more thing. In some scenes, hand-drawn characters are combined with 3DCG models. In this process, the layout is first generated by a rough CG model, and the drawing staff draws the characters according to the layout with separate frames. The final composite of the CG and drawing is created in After Effects and then pasted into the Unity space.

The CG animation was created in 3dsMAX at 24fps. These motions were imported into Unity via FBX, but special data interpolation was used because VR requires speeds of 70 fps or higher.

3 Development of looks and effects and interaction part

Texture expressions for normal animation characters and mechas are constructed by outputting multiple materials in pre-rendering,

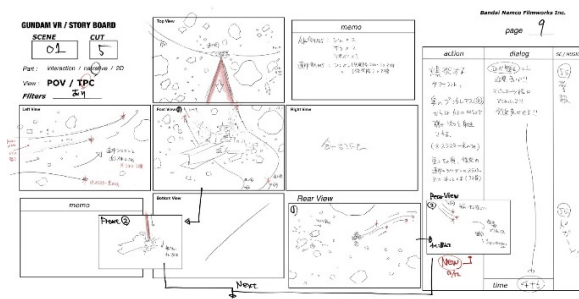


Figure 1: Storyboard format developed for this work.

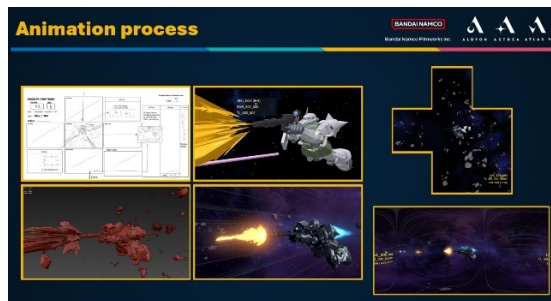


Figure 2: Animation Process Diagram. From storyboard to animatics to finishing touches.



Figure 3: 360-degree development of the Sakuga drawing scene. Unique drawing method never seen before.

but the real-time engine required ingenuity because of the single material and single light.

Another difficulty in specifying colors for drawing was that high-light, normal, and shadow colors behaved differently for different parts. For this reason, we used a look-up table filter for each object as a palette, taking information from the color chart. Encoding the colors on the vertices also reduced memory consumption and improved efficiency.

For lighting, only one basic light was used, but mesh normals were utilized so that lighting could be fine-tuned for each character as needed. This allows detailed lighting adjustments by animating the lighting for each shot.

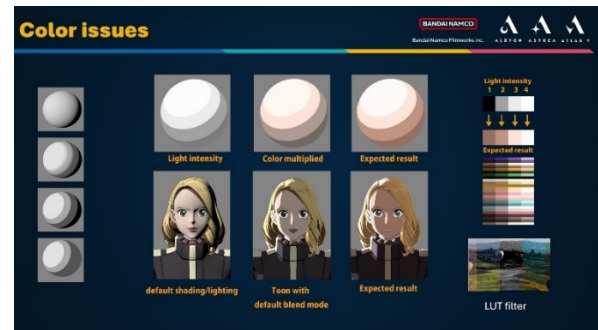


Figure 4: Create small palette textures from color charts to accurately reproduce Anime color specifications.

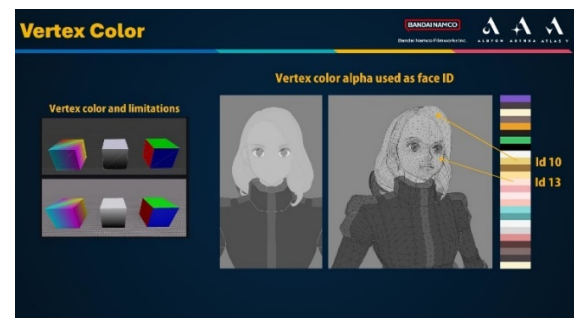


Figure 5: Vertex Color. Vertex color alpha was used as a method to identify color zones.

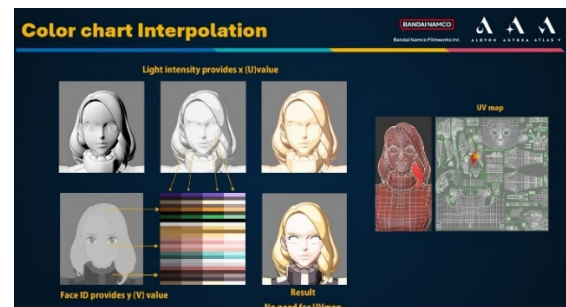


Figure 6: Color Chart Interpolation. The X-axis acquires light intensity and the Y-axis provides the face ID.

This document is up to two pages, so we will not go into detail, but in addition to this, we have created several tools to address various technical issues.

Acknowledgments

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The WIP at the Annecy International Film Festival and the XR nomination at the Venice International Film Festival also gave us confidence. We would also like to take this opportunity to thank all those who have supported us in this project.